The Gender Gaps in Household Tasks and Child Care Over Time

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Introduction

- **Research Question**
  - Within heterosexual two-parent households with at least one child under age 18, how have the gender gaps in time spent on unpaid household tasks and primary child care changed between 2010 and 2019?
- **Background**
  - The gender gaps in time allocation for both paid and unpaid work decreased from 2000 to 2010 across 24 countries, though the gender gaps in unpaid work and child care time persist. (Iza WoL)
- **Motivation**
  - Women continue to bear a disproportionate share of unpaid household work and child care despite workplace gender equality progress.
  - Understanding changes in these gaps over time can inform policy decisions.
  - Medium-run time use trends in the 2010s remain understudied.
- **Data Sources**
  - ATUS respondent, activity, roster, and CPS datasets from 2010-2019
- **Data Manipulation**
  - First, I filtered the ATUS datasets to only include parents within 2-parent heterosexual households with at least one child under age 18.
  - Second, I merged the 40 datasets together for analysis.
  - Final dataset dimensions: 28,132 x 15

Outcome Variables

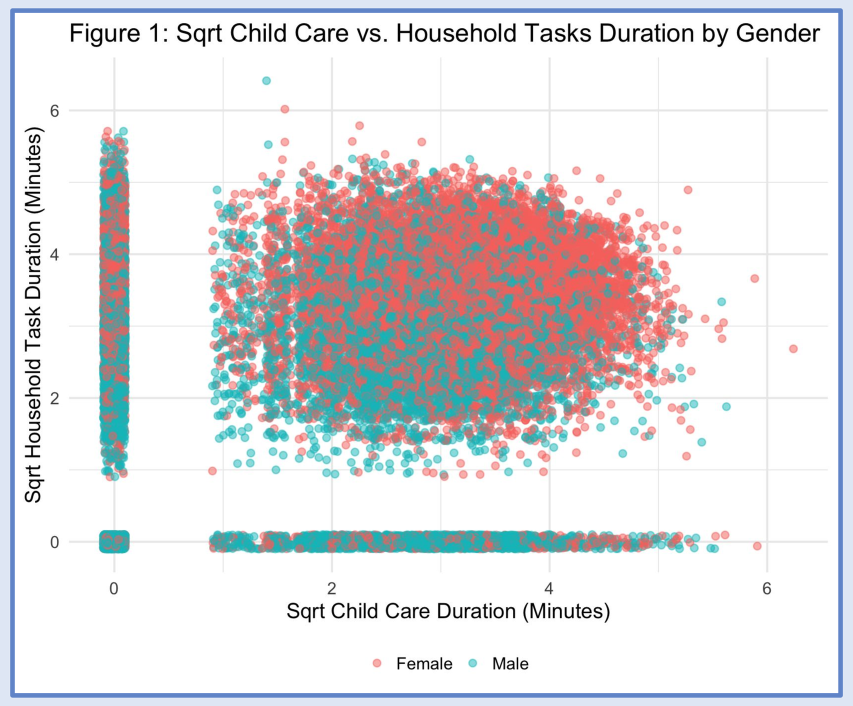

Figure 1: Relationship between sqrt child care and household task duration, 2010 - 2019. Points represent individuals (jittered). Dense bands along the axes indicate respondents who spent zero time on household tasks or child care.

Results: Household Tasks

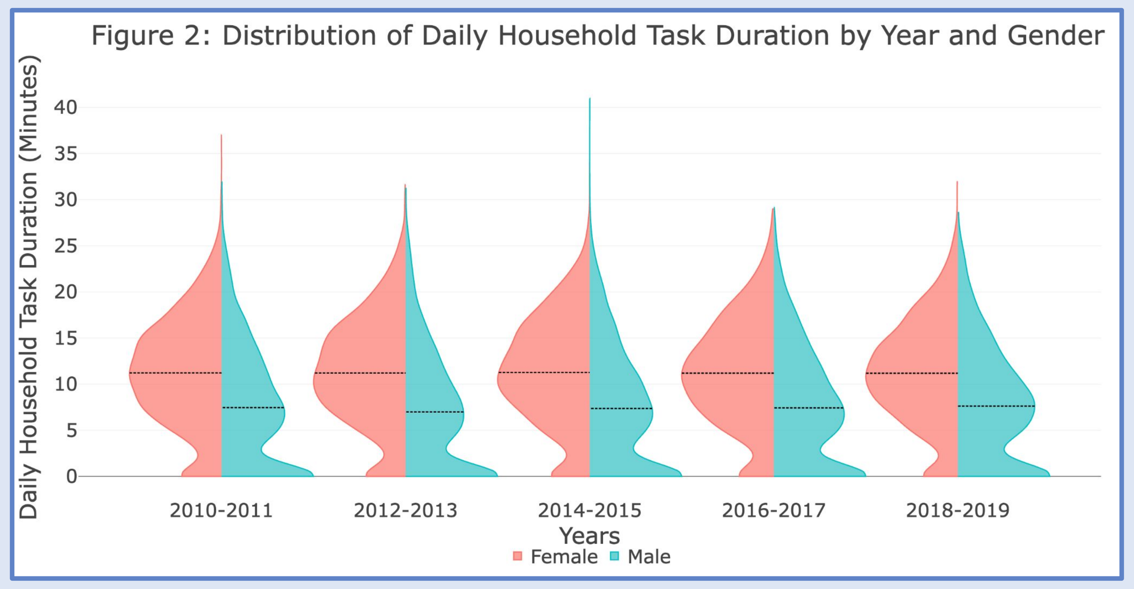

Figure 2: Distribution of daily time spent on household tasks by gender across two-year periods. Split violins show women (left) and men (right) for each period, with a black line indicating the average duration.

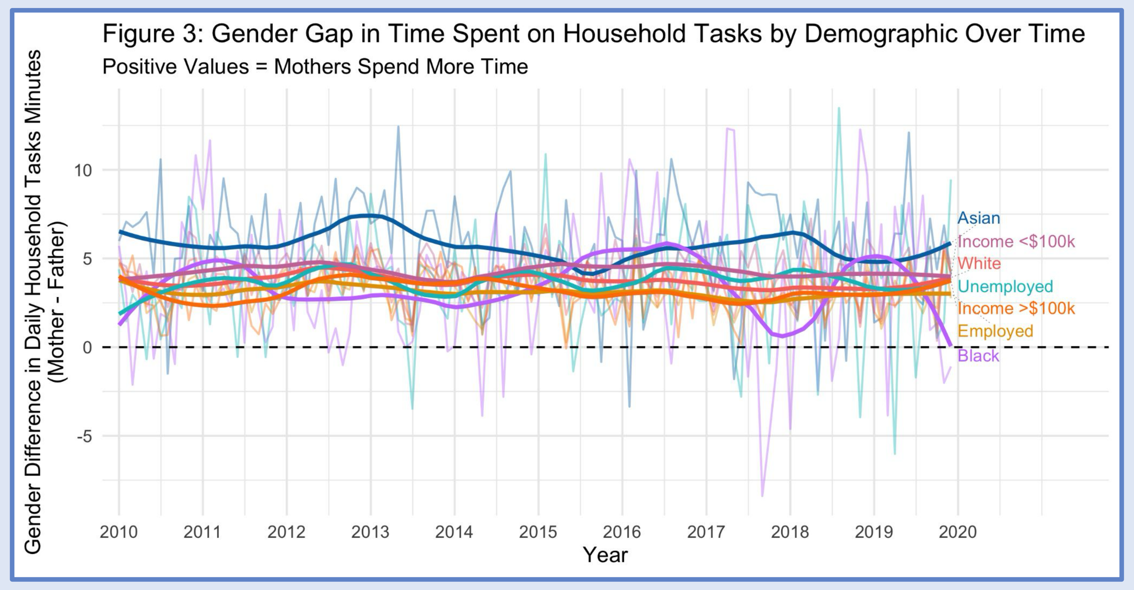

Figure 3: Gender gap in daily time spent on household tasks, broken down by main respondents' race, employment status, and income. Faded lines show raw monthly averages and bolded lines show LOESS-smoothed trends. Unemployed population includes all individuals not actively working for pay.

Results: Child Care

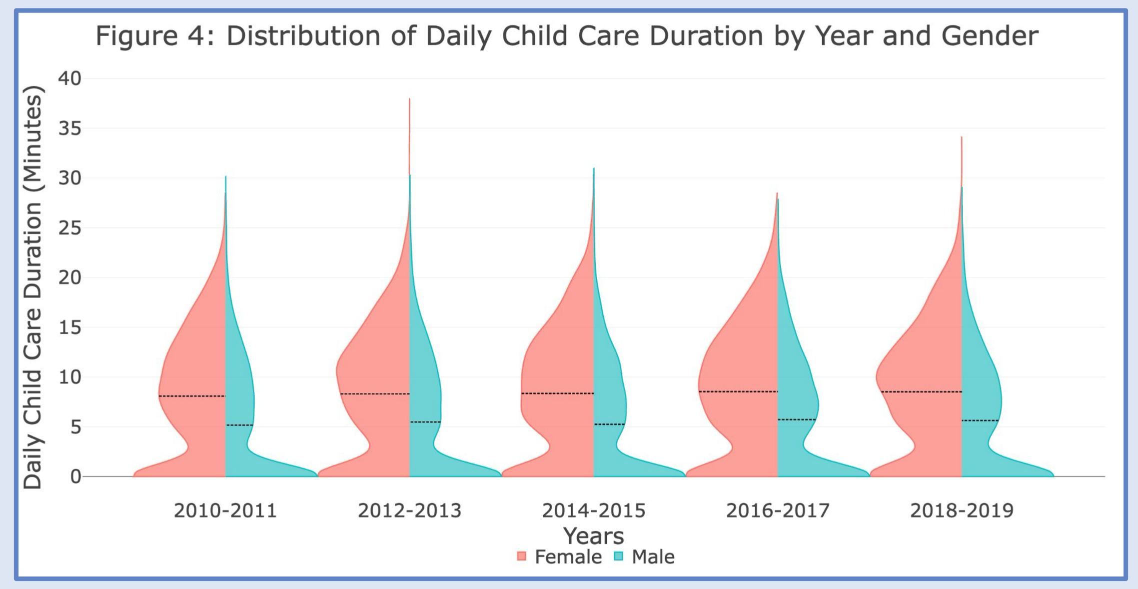

Figure 4: Distribution of daily time spent on child care by gender across two-year periods. Split violins show women (left) and men (right) for each period, with a black line indicating the average duration.

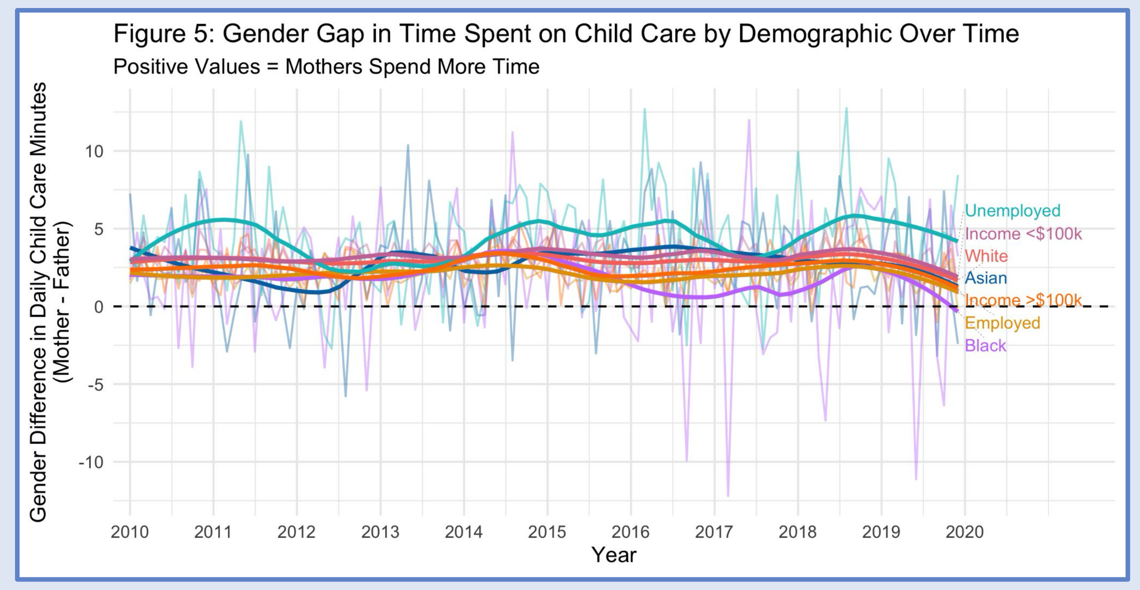

Figure 5: Gender gap in daily time spent on child care, broken down by main respondents' race, employment status, and income. Faded lines show raw monthly averages and bolded lines show LOESS-smoothed trends. Unemployed population includes all individuals not actively working for pay.

Model Results: Household Tasks & Child Care

Table 1: Relationships Between Gender, Time, and Sqrt Child Care / Household Tasks Duration

	(1)	(2)	(3)	(4)
	Sqrt Household Tasks Duration		Sqrt Child Care Duration	
Male	-22.760*	-1.185***	-21.401	-1.115***
Year	0.004		0.007	
Date		0.00001		0.00002
Male * Year	0.011*		0.010	
Male * Date		0.00003		0.00003
Weekend	0.361***		-0.277***	
Employed	-0.551***	-0.555***	-0.513***	-0.509***
Income Level	0.038***	0.038***	0.104***	0.105***
Asian	-0.059*	-0.064*	0.079**	0.084**
Black	-0.271***	-0.273***	-0.294**	-0.293***
Multiracial	-0.067	-0.048	-0.029	-0.044
Constant	-4.931	3.052***	-12.659	2.094***
Adjusted R-Squared	0.139	0.123	0.088	0.080
Partial F-Test P-Value	< 0.001	< 0.001	< 0.001	< 0.001
RMSE (Train Sample)	1.323	1.337	1.485	1.493
RMSE (Held-Out Test Sample)	1.342	1.359	1.529	1.538

Table 1: The outcome variable in models (1) and (2) is the square root of the time spent completing household tasks per day. The outcome variable in models (3) and (4) is the square root of the time spent on child care per day. Models (1) and (3) include a categorical year variable, while models (2) and (4) include a continuous date variable. The interaction terms Male * Year and Male * Date capture whether the gender gap in time spent on child care and household tasks changed over time. Square root transformations are applied to reduce right skews in the outcome variables, making the model results robust to deviations from the normality assumption. A categorical state variable is also included in the models. The models are estimated on a randomly selected training sample comprising 85% of observations (n = 23,912) and evaluated on a held-out test sample comprising the remaining 15% of observations (n = 4,220). The Partial F-Test P-Value row reports results from ANOVA comparisons between each full model and a restricted model excluding gender, time, and their interaction; a significant p-value indicates that the terms jointly improve model fit. The RMSE rows use either the train sample or the held-out test sample, and are reported in units of square root minutes. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

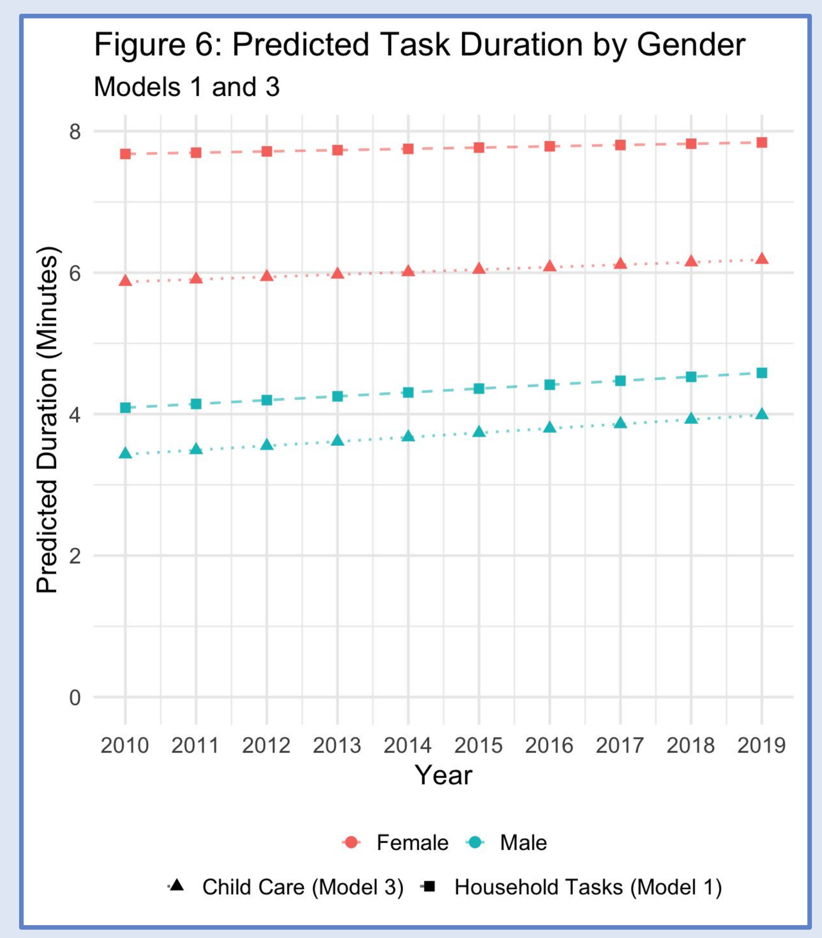

Figure 6: Predicted number of minutes spent on household tasks and child care by gender, based on Models 1 and 3. Predictions are generated by holding all other covariates at their most common values: weekday, employed, income $50,000 - $99,999, White, California, and median household task (3.08 minutes) or child care (2.59 minutes) duration. Predicted minutes are obtained by squaring the model's sqrt-scale predictions.

Conclusion

- **Overall**
  - Gender alone is a meaningful predictor of time spent on household tasks and child care, but the interaction between gender and time is not significant.
  - The gender gaps in time spent on household tasks and child care remained relatively steady throughout the 2010s, but the gaps are negligible.
- **Household Tasks**
  - On average, women spend 3.80 more minutes daily on household tasks than men, controlling for covariates.
  - The largest gender gap over time occurs for Asian main respondents.
- **Child Care**
  - On average, women spend 2.26 more minutes daily on child care than men, controlling for covariates.
  - The largest gender gap over time occurs for unemployed main respondents.
  - There is a slight narrowing of the gender gap beginning in mid-2018.

Data Ethics & Limitations

- **Data Ethics**
  - Ethical data: anonymized dataset from the publically available ATUS study administered by the US Bureau of Labor Statistics
  - Race: race-based findings should be interpreted within the context of broader structural inequities
- **Limitations**
  - Household structures: this study focuses on two-parent heterosexual households (which excludes many family structures), so findings should not be generalized beyond these family compositions
  - Reporting bias: respondents may over- or under-report time spent on certain activities that may differ systematically by gender